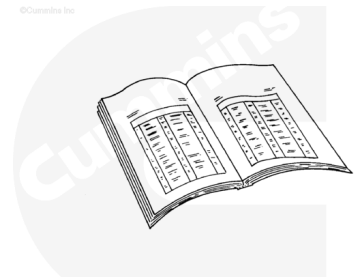


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



ck800wa

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries. Refer to the original equipment manufacturer (OEM) service manual.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/269/269-008-018-tr.html>)
- Remove the air piping from the intake manifold and turbocharger. Refer to the OEM service manual.
- Remove the crankcase breather. Refer to Procedure 003-001 in Section 3. (</qs3/pubsys2/xml/en/procedures/269/269-003-001-tr.html>)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/269/269-003-011-tr.html>)

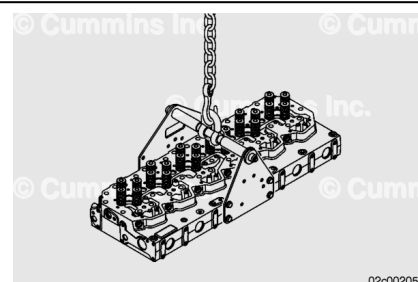
- Remove the engine brakes, if applicable. Refer to Procedure 020-024 in Section 20.
(/qs3/pubsys2/xml/en/procedures/269/269-020-024-tr.html)
- Remove the push rods or push tubes. Refer to Procedure 004-014 in Section 4.
(/qs3/pubsys2/xml/en/procedures/269/269-004-014-tr.html)
- Remove the rocker lever assemblies. Refer to Procedure 003-009 in Section 3.
(/qs3/pubsys2/xml/en/procedures/269/269-003-009-tr.html)
- Remove the internal actuator harness. Refer to Procedure 019-063 in Section 19.
(/qs3/pubsys2/xml/en/procedures/269/269-019-063-tr.html)
- Remove the crossheads. Refer to Procedure 002-001 in Section 2. (/qs3/pubsys2/xml/en/procedures/269/269-002-001-tr.html)
- Remove the high-pressure injector supply lines. Refer to Procedure 006-051 in Section 6.
(/qs3/pubsys2/xml/en/procedures/269/269-006-051-tr.html)
- Remove the fuel rail. Refer to Procedure 006-060 in Section 6. (/qs3/pubsys2/xml/en/procedures/269/269-006-060-tr.html)
- Remove the fuel connectors. Refer to Procedure 006-052 in Section 6. (/qs3/pubsys2/xml/en/procedures/269/269-006-052-tr.html)
- Remove the injectors. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/269/269-006-026-tr.html)
- Remove the cooling thermostat housing. Refer to Procedure 008-013 in Section 8.
(/qs3/pubsys2/xml/en/procedures/269/269-008-013.html)
- Remove the air compressor water outlet tube. Refer to Procedure 012-004 in Section 12.
(/qs3/pubsys2/xml/en/procedures/269/269-012-004-tr.html)
- Remove the rocker lever housing. Refer to Procedure 003-013 in Section 3.
(/qs3/pubsys2/xml/en/procedures/269/269-003-013-tr.html)

- Remove the turbocharger. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/269/269-010-033-tr.html>)
- Remove the exhaust manifold. Refer to Procedure 011-007 in Section 11. (</qs3/pubsys2/xml/en/procedures/269/269-011-007-tr.html>)
- Remove the fuel drain line. Refer to Procedure 006-013 in Section 6. (</qs3/pubsys2/xml/en/procedures/269/269-006-013-tr.html>)
- Remove the air intake manifold. Refer to Procedure 010-023 in Section 10. (</qs3/pubsys2/xml/en/procedures/269/269-010-023-tr.html>)
- Remove the engine lifting brackets. Refer to Procedure 016-001 in Section 16. (</qs3/pubsys2/xml/en/procedures/269/269-016-001-tr.html>)
- Remove the fan hub, if applicable. Refer to Procedure 008-036 in Section 8. (</qs3/pubsys2/xml/en/procedures/269/269-008-036-tr.html>)
- Remove all the necessary brackets and clamps.

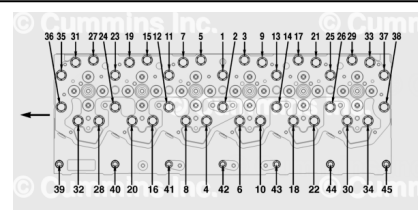
Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



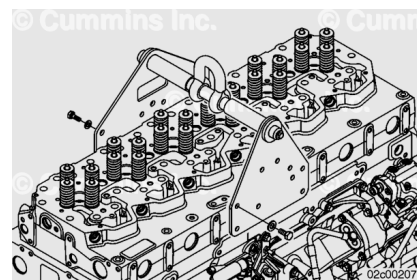
Remove the cylinder head capscrews (38 M15 and 7 M10) in the sequence shown.



Install cylinder head lifting bracket, Part Number 4919361. Use the eight M10 capscrews that are included in the cylinder head lifting fixture kit, Kit Number 4919360.

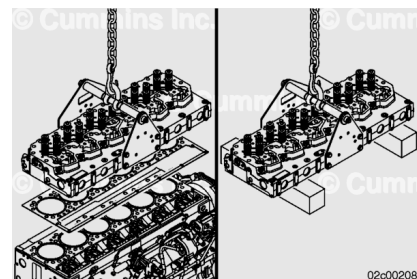
Tighten the capscrews.

Torque Value: 34 n•m [25 ft-lb]



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

Place the cylinder head on wood blocks to prevent damage to the combustion face.

⚠ CAUTION ⚠

Immediately upon removal of the cylinder head, plugs must be installed into all of the oil passage drillings in both the cylinder head and the cylinder block. The plugs are necessary to prevent debris from entering the lubricating system during the repair. Failure to insert oil passage plugs can result in connecting rod bearing failure, crankshaft failure, or both.

Use a hoist or hydraulic arm to remove the cylinder head.

Remove the cylinder head gasket.

Reusable plugs, Part Number 4089255, are included in the master plug kit, Part Number 4918319.

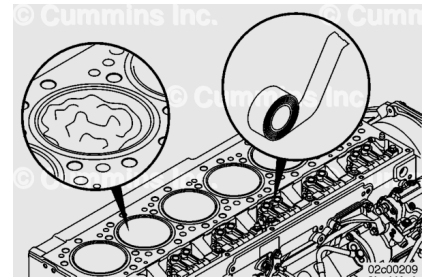
Install the plugs in the cylinder block and cylinder head oil drillings.

Clean

Place clean, lint-free rags above the pistons to prevent debris from falling into the engine.

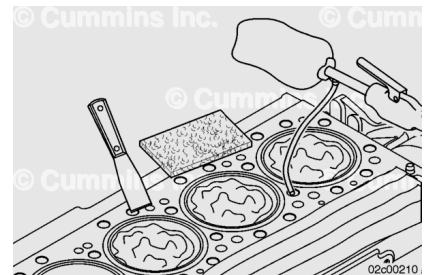
Place tape over the camshaft and cam follower ports to prevent debris from falling into the engine.

Plug or cover the coolant and oil passages in the cylinder block deck.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use a gasket scraper to clean the cylinder block deck surface.

Use Scotch-Brite™ 7448 abrasive pad, Part Number 3823258 or equivalent, and solvent to remove any residual gasket material from the cylinder block deck surface.

Use a chip vacuum to make sure the cylinder head capscrew holes are clean and free of debris, oil, and coolant.

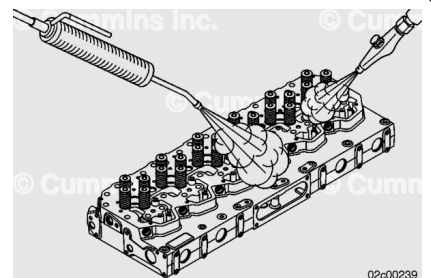
Do **not** use power tools to clean the cylinder block head deck.

Use the following procedure to inspect the cylinder block head deck. Refer to Procedure 001-026 in Section 1.

(/qs3/pubsys2/xml/en/procedures/269/269-001-026.html)

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

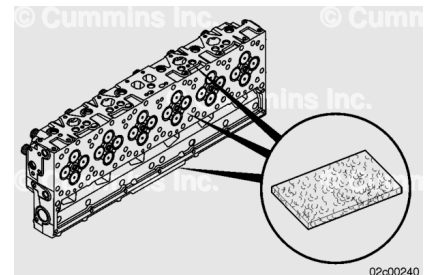
Steam clean the cylinder head.

Dry with compressed air.

Use compressed air to blow out all capscrew holes.

⚠ WARNING ⚠

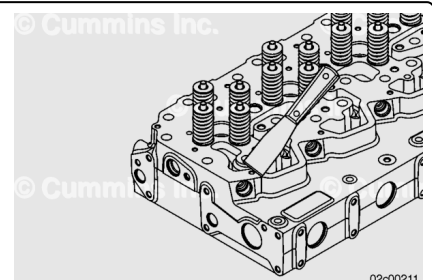
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendation for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Use Scotch-Brite™ 7448 abrasive pad, Part Number 3823258 or equivalent, and solvent to clean the cylinder head combustion face.

Clean the exhaust manifold and intake manifold gasket surfaces.

Clean the rocker housing gasket surface of any debris. Refer to Procedure 003-013 in Section 3.
(/qs3/pubsys2/xml/en/procedures/269/269-003-013-tr.html)

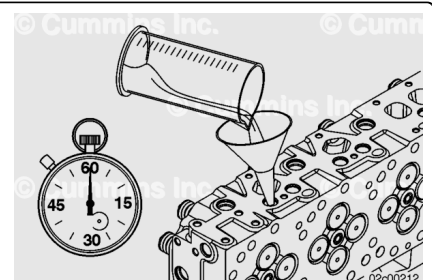


Leak Test

Inspect the valves for indications of leaking or burning. If indications of leaking or burning are found, the valves and the seats **must** be resurfaced.

Test the cylinder head for damage.

Set the cylinder head down with the exhaust ports facing up. Secure the cylinder head, as it may tip over due to the thinner intake face.



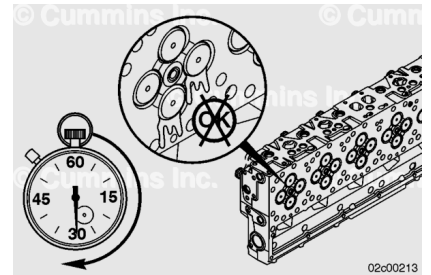
Pour fuel into one of the exhaust ports until it is full. Set the container of fuel down, and start a timer.

If a drop of fuel drips or runs down the face of the cylinder head within 30 seconds, the exhaust valves and the seats **must** be resurfaced.

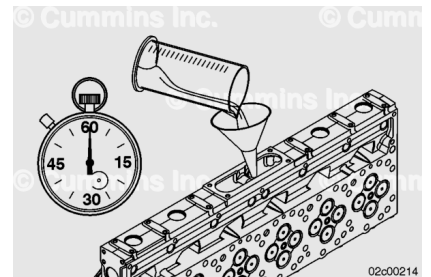
If a drop of fuel has **not** run down the face of the cylinder head in 30 seconds, the exhaust valves are acceptable.

Repeat the process for all six cylinders.

Flip the cylinder head over and set it down with the intake ports facing up. Secure the cylinder head to prevent it from tipping over, due to the thinner intake face.



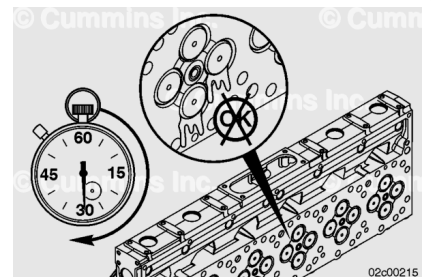
Pour fuel into one of the intake ports until it is full. Set the container of fuel down, and start the timer.



If a drop of fuel drips or runs down the face of the cylinder head within 30 seconds, the intake valves and the seats **must** be resurfaced.

If a drop of fuel has **not** run down the face of the cylinder head in 30 seconds, the intake valves are acceptable.

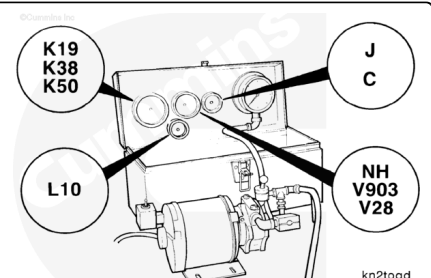
If any of the intake or exhaust valves do **not** pass the test, all of the valves and valve seats **must** be resurfaced.



Vacuum Test

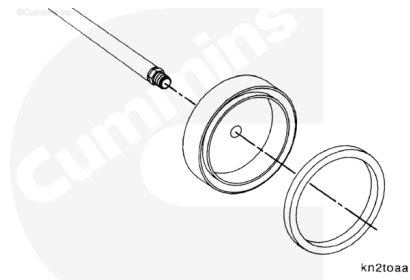
Use the vacuum tester, Part Number ST-1257, to inspect the seal between the valve and the valve seat.

The valve vacuum tester is **not** recommended for use on used cylinder heads. See the Leak Test section of this procedure for the recommended procedure to check used cylinder heads.



The valve vacuum tester can be used to test all Cummins® engine models. The seal ring, Part Number ST-1257-35, and vacuum cup, Part Number 3376100, are used on the ISZ13 cylinder heads.

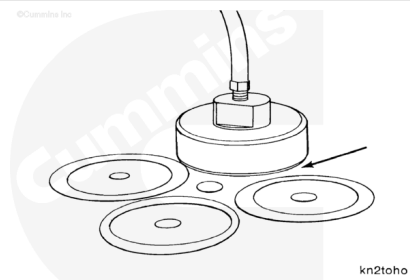
Install the seal ring and vacuum cup to the vacuum line hose.



The valves and valve seats **must** be clean and dry when vacuum testing.

Cover the valve with the cup and seal.

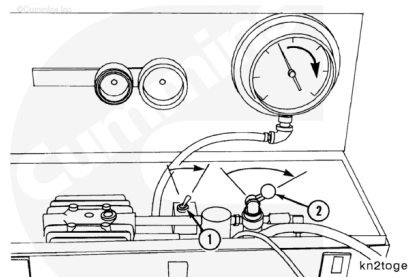
The seal **must** make a tight contact on the cylinder head around the valve.



Move the toggle switch (1) to the ON position.

Turn the vacuum control valve (2) to the OPEN position.

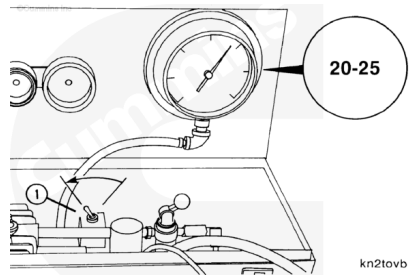
The vacuum control valve is in the OPEN position if the vacuum gauge needle moves **clockwise**.



Operate the vacuum pump until the gauge indicates the specified vacuum.

Valve to Valve Seat Vacuum

mm-hg		in-hg
508	MIN	20
685	MAX	27

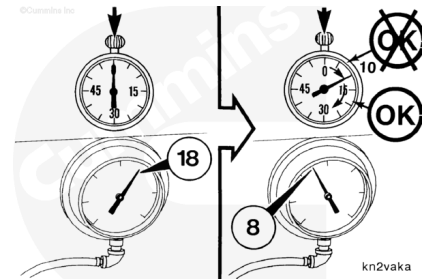


Turn the toggle switch to the OFF position.

Use a stopwatch and start timing when the needle on the gauge indicates 457 mm Hg [18 in Hg].

Stop timing when the needle on the gauge indicates 203 mm Hg [8 in Hg].

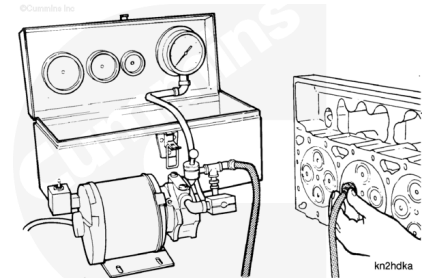
The elapsed time for the needle to move between the specified gauge readings **must** be 10 seconds or more.



If the elapsed time is less than 10 seconds, perform the following checks:

- Repeat the test to be sure the equipment is operating properly.
- Use a mallet to hit the valve stem lightly to make sure the valve is seated. Repeat the test.
- Apply a thin layer of grease on the outside diameters of the insert and the valve head. Repeat the test. The grease pattern will show the point of leakage.

A break in the grease seal pattern will indicate leakage between the valves and valve seat or the valve seat insert and the cylinder head.



Disassemble

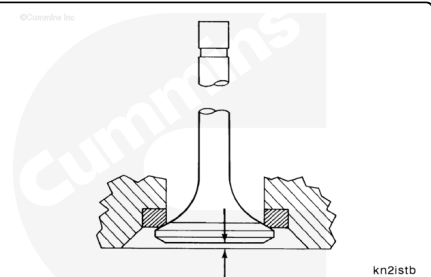
Note : Injector sleeves are **not** serviceable. Special tools and precision machining are required to replace the injector sleeves. If Cummins Inc. approved tools and procedures are **not** available to replace the injector sleeves, the cylinder head **must** be replaced.

Perform the coolant passage pressure test to determine if there is an injector sleeve leak. Use the Pressure Test section of this procedure to perform the pressure test. Replace the cylinder head if any of the injector sleeves leak.



Prior to disassembly, measure and record the valve recess. Use the Clean and Inspect for Reuse section in this procedure.

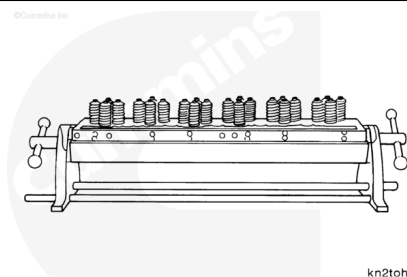
Make sure the measurement tool is flat and flush on the head surface. Measure the flat surface of the valve, avoiding the scalloped areas.



Maximum Operational Valve Recession

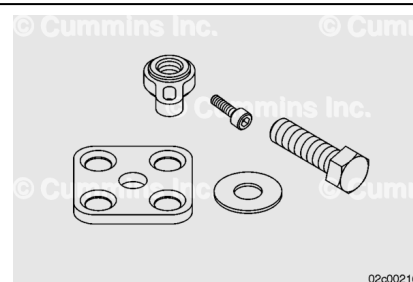
mm		in
3.05	MAX	0.120

Install the cylinder head in the head holding fixture, Part Number ST-583, as illustrated.



⚠ WARNING ⚠

Valve springs are under tension and can act as projectiles if improperly released. To reduce the possibility of eye injury, wear safety glasses with side shields. Before releasing the valve spring tension, check that the collets are properly seated.



⚠ CAUTION ⚠

Do not use an impact tool for any part of this procedure. Doing so can damage the tool and/or the engine.

Use the four valve spring compressor, Part Number 3163940, to compress the valve springs.

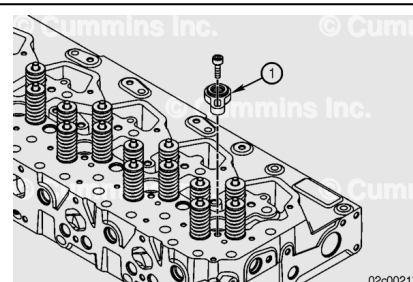
The forcing capscrew pilots into the injector bore and is secured to a mounting plate attached at the injector hold down clamp screw hole.

Install a wood block between the valves and head holding fixture to support the valves.

Secure the mounting plate (1) with the larger hole over the injector bore, such that the mounting plate capscrew hole is over the injector hold down clamp capscrew hole. Install the mounting plate capscrew into the injector hold clamp capscrew hole.

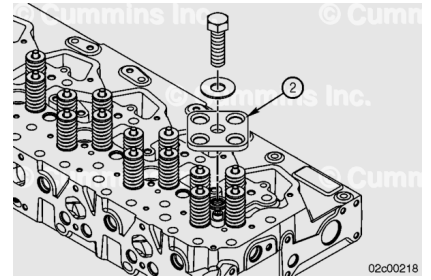
Tighten the capscrew.

Torque Value: 54 n·m [40 ft·lb]



⚠ CAUTION ⚠

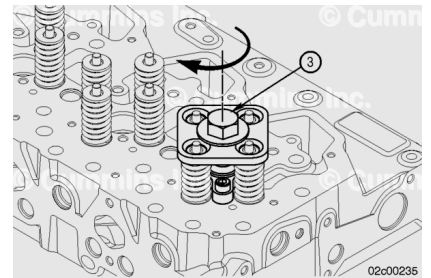
The side of the retainer plate with the larger diameter chamfers faces down to accommodate the valve springs. Failure to position the retainer plate properly can result in damage to the engine and/or tool.



Position the compressor plate (2) over the forcing capscrew and onto the valve springs.

⚠ WARNING ⚠

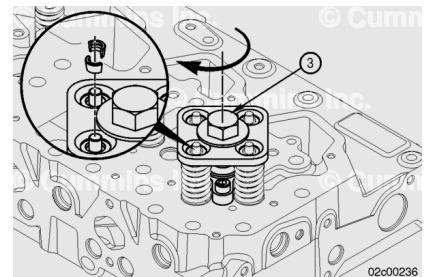
Eye protection should be worn during this procedure. Springs are under tension and can act as a projectile if released, causing serious injury.



Turn the forcing capscrew (3) **clockwise** until the valve springs start to compress.

⚠ CAUTION ⚠

Do not overtighten the forcing nut. Doing so can damage the tool and/or springs.

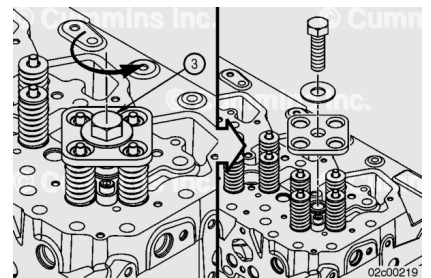


Continue turning the forcing capscrew (3) **clockwise** until the valve collets can be removed with a magnetic tool.

Discard the valve collets.

⚠ WARNING ⚠

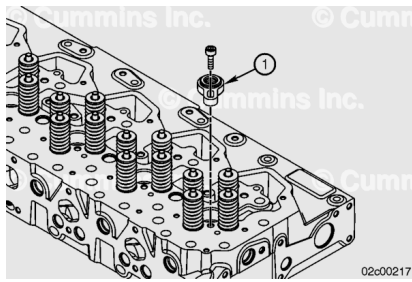
Wear eye protection during this procedure. Springs are under tension and can act as a projectile if released, causing serious injury.



Carefully turn the forcing capscrew (3) **counterclockwise** until the tension is released from the valve springs.

Remove the forcing capscrew and compressor plate.

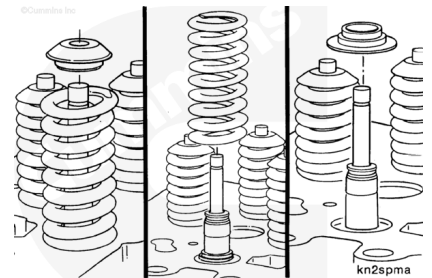
Remove the mounting plate by removing the injector hold-down bolt.



Remove the valve spring retainer.

Remove the valve spring.

Remove the valve spring wear plate.

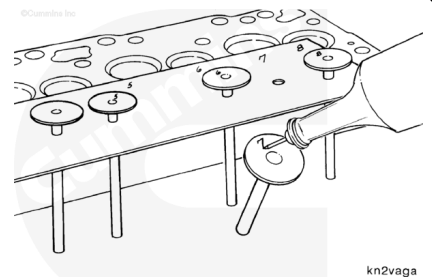


Remove the valve.

Mark the valves with an electric pencil (engraving tool) for location as they are removed. See the illustration for marking location.

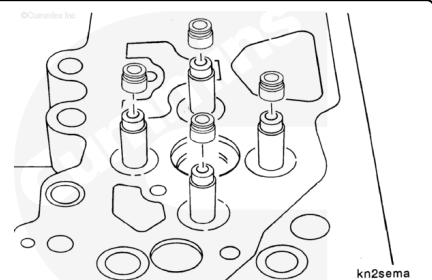
The intake and exhaust valves are manufactured from different materials and are **not** interchangeable.

Repeat the procedure to remove the remaining valve collets, retainers, springs, wear plates, and valves.

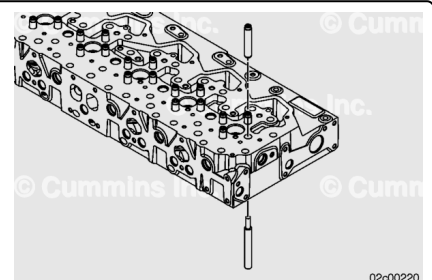


Remove the valve seals from the valve guides.

Prior to removing the valve guides, measure and record the valve guide inside diameter and height. Use the Clean and Inspect for Reuse section of this procedure.

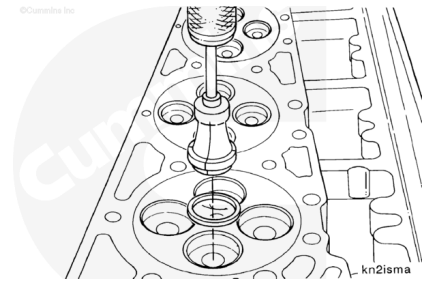


Use valve guide driver, Part Number 3823271, to remove the old valve guides.



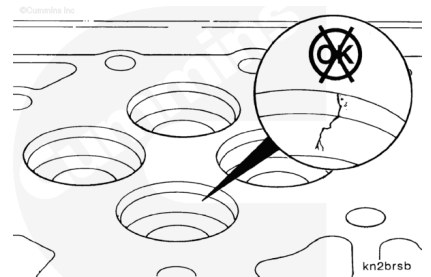
Prior to removing the valve seat inserts, see the Leak Test and the Inspect for Reuse sections of this procedure. The condition of the valve, the amount of recess, and the sealing of the valve on the seat insert all help determine whether or **not** a seat insert needs to be replaced.

Use slide hammer assembly, Part Number 3376616, and valve seat extractor, Part Number 3376146, to remove the valve seat inserts from the cylinder head.



Inspect the insert bore for cracks or other damage.

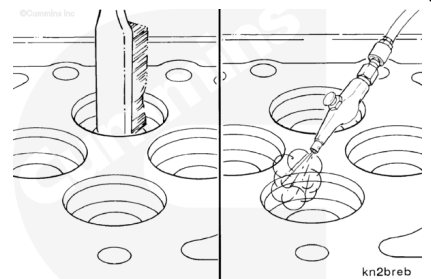
If cracks or other damage are found, the cylinder head **must** be replaced.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use a wire brush and solvent to clean the deposits from the valve seat insert bores.

Use solvent to clean the cylinder head.

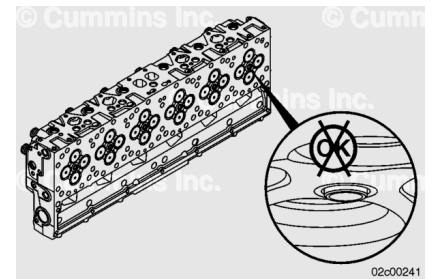
Dry with compressed air.

Use solvent to clean the injector bore.

Dry with compressed air.

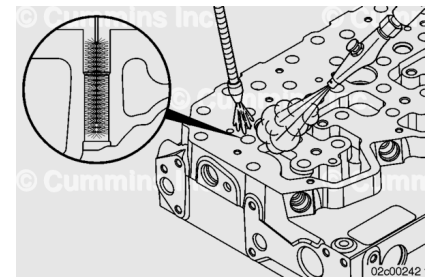
Inspect the injector seat surface at the bottom of the cylinder head.

If the injector sleeve protrudes past the head deck, replace the cylinder head.



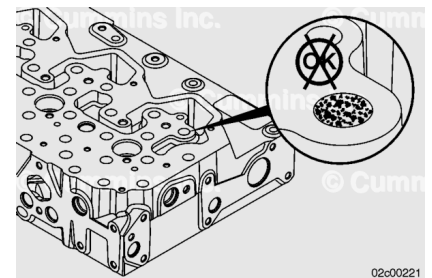
Use a flexible brush and solvent to clean the valve guide bores in the cylinder head.

Dry with compressed air.



Inspect the water passages for restrictions or foreign material.

Remove any obstructions.

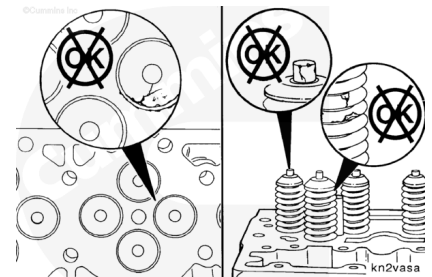


Inspect the valves and valve springs for cracks, bent or broken valve stems, broken valve spring collets, or other damage.

Inspect the valves for indications of leaking or burning.

If any of the following damage is found, the cylinder head **must** be rebuilt or replaced:

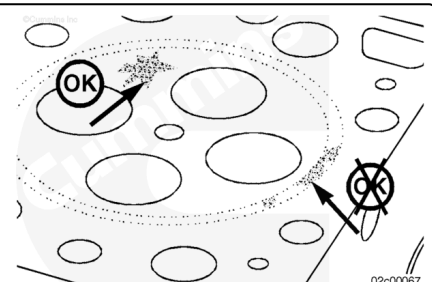
- Cracked
- Damaged
- Leaking
- Burned valves.



Inspect the cylinder head and valves for cracks or other damage.

If a crack in the cylinder head is suspected, pressure-test the cylinder head. See the Pressure Test section of this procedure.

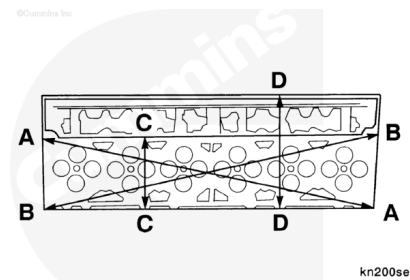
Inspect the combustion face for ring-shaped wear or erosion that encircles the valves.



Measure the flatness of the cylinder head combustion face surface as follows:

- AA and BB (corner to corner)
- CC (across combustion face)
- DD (across entire cylinder head surface).

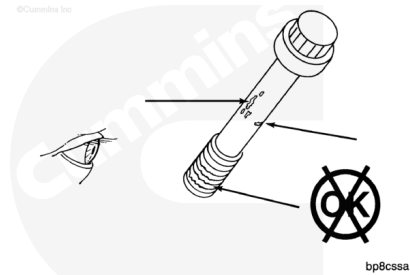
Dimensions CC and DD **must** be checked from front to rear of cylinder head.



Cylinder Head Flatness				
	mm		in	
AA and BB		0.200	MAX	0.008
CC		0.076	MAX	0.003
DD		0.127	MAX	0.005

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendation for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ CAUTION ⚠

Do not use caustic or acid solutions to clean the cylinder head capscrews to avoid damage to cylinder head capscrews.

Clean the cylinder head capscrews with a petroleum-based solvent.

Clean the capscrews thoroughly with a wire brush, a wire wheel (soft), or a nonabrasive bead blast to remove deposits from the shank and threads.

Inspect the capscrews for the following:

- Damaged threads
- Rust or corrosion caused by pitting
- Nicks, bent, stretched, or galled.

The capscrew must be replaced if it has any of the listed damages. Refer to Procedure 017-001 in Section 17.

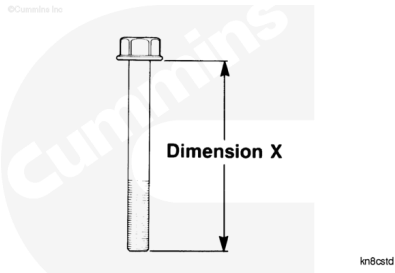
(/qs3/pubsys2/xml/en/procedures/269/269-017-001.html)

The capscrew can be checked using a set of calipers. The maximum allowable free length is measured from the bottom of the flange to the end of the capscrew (dimension X).

Cylinder Head Capscrew Free Length (M15 Hexagon Head)

mm		in
197.00	MAX	7.756

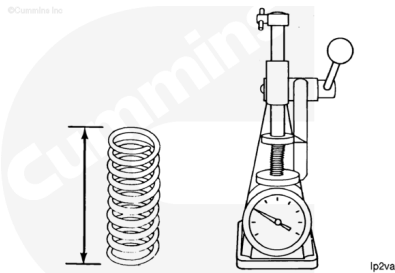
Note : M10 hexagon head capscrew measurements are not required.



Use valve spring tester, Part Number 3375182, to measure the valve spring load at the valve spring working height.

Valve Spring Load at 44.56 mm [1.754 in]

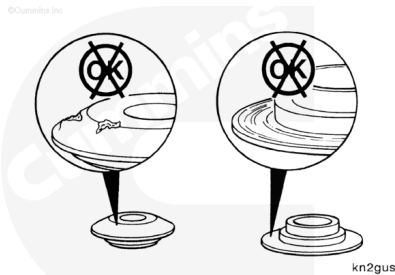
n		lbf
1000	MIN	225
1265	MAX	284



If the valve spring load for the working height is less than the minimum specified, the valve spring **must** be replaced.

Inspect the valve spring retainers and valve spring guides for damage or work areas.

Discard damaged and worn parts.



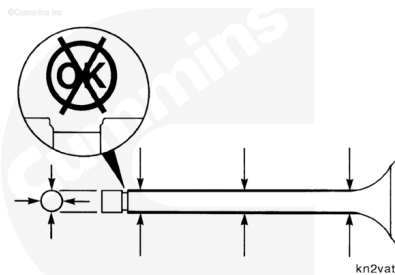
Inspect the valves for damage.

Inspect the collet grooves for wear.

Measure the outside diameter of the valve stem.

Valve Stem Outside Diameter

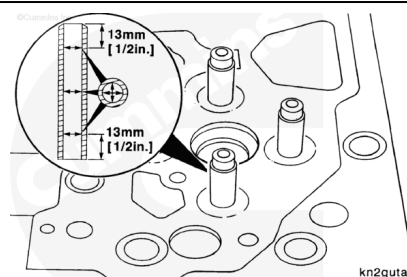
mm		in
9.580	MIN	0.3772
9.633	MAX	0.3793



If the valves are damaged or the stems are worn smaller than the minimum specified, the valves **must** be replaced.

Inspect the valve guides, if **not** removed in the Disassemble step, for chips or cracks.

Measure the valve guide inside diameter.

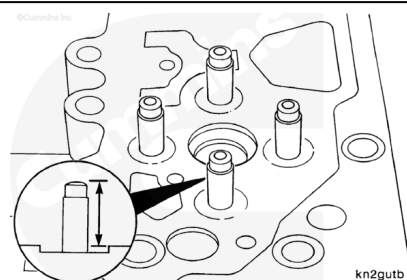


New Valve Guide Inside Diameter, Installed		
mm		in
9.670	MIN	0.3807
9.695	MAX	0.3817

If the valve guide is **not** within the specifications given, the valve guide **must** be replaced.

Note : If the valve guides have been removed, the valve guides **must** be replaced with new valve guides.

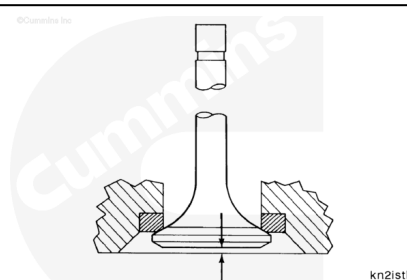
Measure the valve guide installed height above the cylinder head top deck surface.



Valve Guide Height (Installed)		
mm		in
24.62	MIN	0.969
24.88	MAX	0.979

Install a new valve in the valve guide. Hold the valve firmly against the valve seat insert.

Use a depth gauge, Part Number 3823495, to measure the valve recess in the cylinder head.



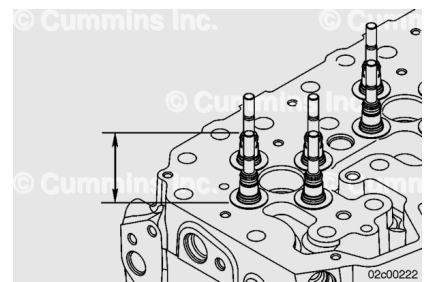
Valve Recession			
	mm		in
Intake	1.003	MIN	0.040
	1.384	MAX	0.055
Exhaust	0.953	MIN	0.038
	1.334	MAX	0.053

Maximum Valve Recession Variation
Between Valves on Same Cylinder

	mm		in	
Intake		1.003	MAX	0.040
Exhaust		0.953	MAX	0.038

Maximum Valve Tip Height

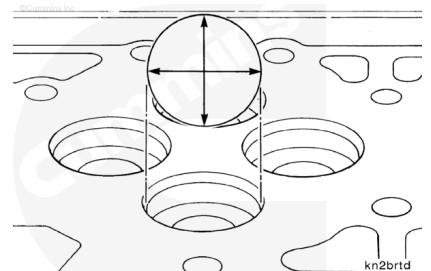
mm		in
99.60	MAX	3.921



Measure the inside diameter of the valve seat insert bore in the cylinder head.

Insert Bore Inside Diameter

	mm		in
Intake	46.51	MIN	1.831
	46.55	MAX	1.833
Exhaust	44.51	MIN	1.752
	44.55	MAX	1.754

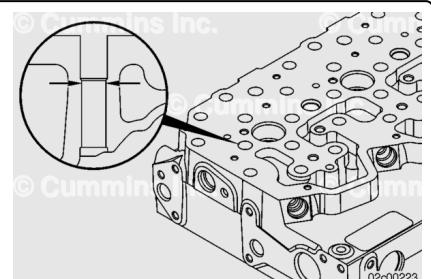


Measure the inside diameter of the valve guide bore in the cylinder head.

Make sure the measurement is taken in the pressed area (smaller diameter below the step).

Valve Guide Bore Inside Diameter

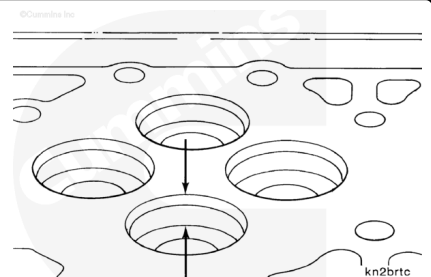
mm		in
16.477	MIN	0.6475
16.503	MAX	0.6497



Measure the valve seat insert bore depth in the cylinder head.

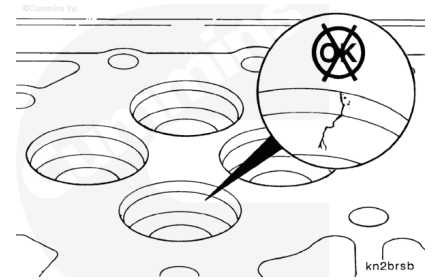
Insert Bore Depth (Standard Insert)

	mm		in
Intake	9.32	MIN	0.367
	9.48	MAX	0.373
Exhaust	10.32	MIN	0.406
	10.48	MAX	0.413



Inspect the insert bore for cracks or other damage.

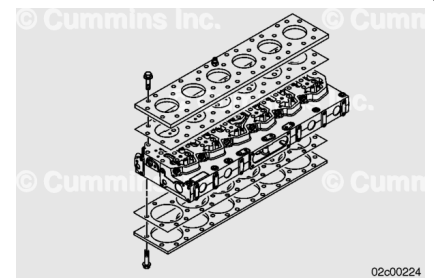
If cracks or other damage is found, the cylinder head **must** be replaced.



Pressure Test

⚠ CAUTION ⚠

Do not pressure-test the cylinder head with the valves and valve springs installed. Water entering the cylinder head can not be dried thoroughly and will damage the valve guides and valve stems.



Install cylinder head water test fixture, Part Number 4919065.

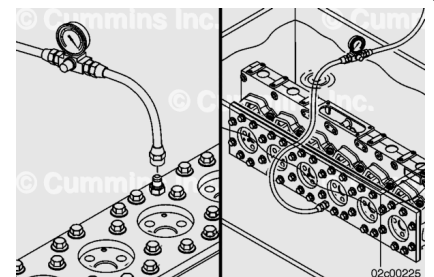
- Install gaskets, Part Numbers 4919068 and 4919069
- Install upper test plate, Part Number 4919066
- Install lower test plate, Part Number 4919067
- Install the 20 capscrews and nuts included with kit, Part number 4919065.

Tighten the nuts.

Torque Value: 215 n·m [159 ft·lb]

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Connect a regulated air supply hose to the test fixture plate.
Apply air pressure.

Measurements

	kpa	psi
Pressure Test Air Pressure	138	20

Use a nylon lifting strap and a hoist to place the cylinder head in a tank of heated water.

Completely submerge the cylinder head in the water.

Measurements

	celsius	fahrenheit
Test Tank Water Temperature	60	140

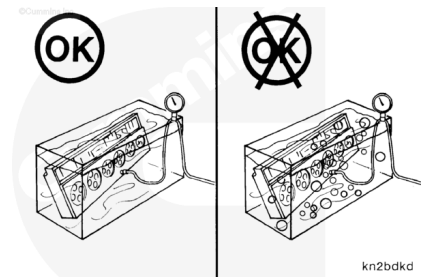
Inspect for air bubbles rising from the water.

If air bubbles are seen, replace or rebuild the cylinder head.

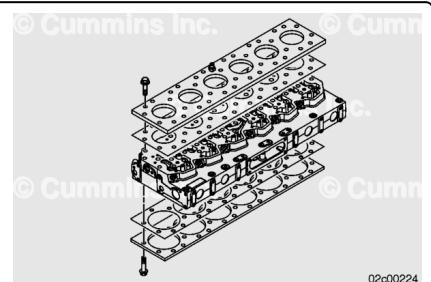
If a crack is found in the cylinder head, preserve the cylinder head with clean engine oil so that rust does **not** develop.

If the source of the air bubbles are coming from a cast portion of the cylinder head or injector cup area, the cylinder head **must** be replaced.

If the source of the air bubbles are from permanent assembly components, such as cup plugs, the cylinder head can be repaired.



Remove the test equipment.

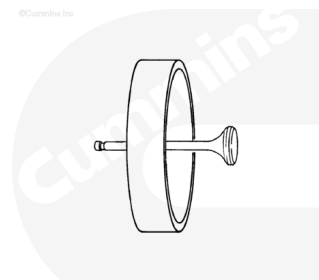


Magnetic Crack Inspect

Use the magnetic particle residual method to inspect the valves for cracks.

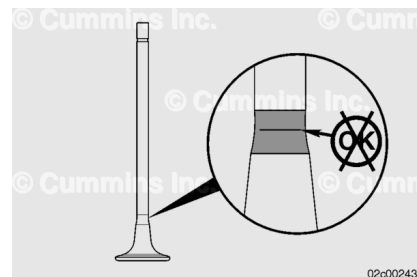
The exhaust valves contain two types of metal and **must** be inspected by the coil shot method. There will be a magnetic leakage apparent at the point where the two metals are welded together. The leakage will appear as a wide pattern of magnetic particles.

Magnetize the valves in a coil of 100 to 200 amperes.



Inspect the valves with residual Magnaglo®.

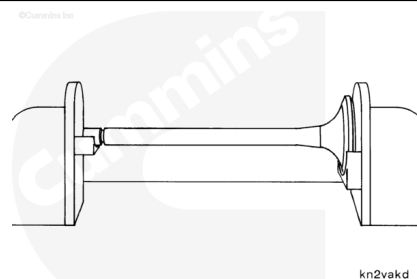
A broad fuzzy pattern will appear at the welded joint on the exhaust valves. If there is a distinct line in the pattern, the valve **must** be replaced.



The intake valves contain **only** one type of metal and **must** be magnetized and inspected in two directions. Use the Magnaglo® coil method and inspect with residual Magnaglo®, the same as for the exhaust valves.

Magnetize the intake valves again with a headshot at 500 to 700 amperes.

Inspect the valves again with residual Magnaglo®.

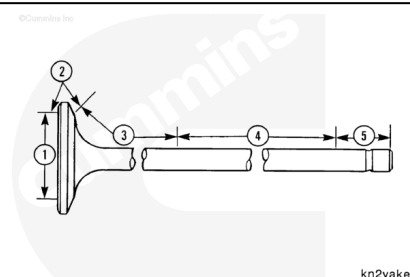


Acceptance Criteria (Exhaust and Intake Valves)

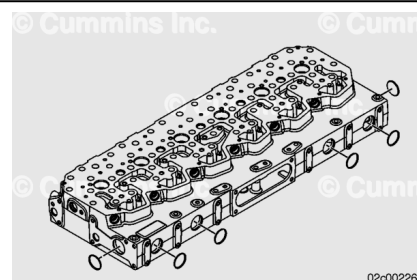
A visible crack is one which can be seen through a three power magnifying glass after the magnetic particle suspension is removed.

The valve **must not** have any magnetic indications or visible indication.

Remove all magnetism and clean the acceptable valves.

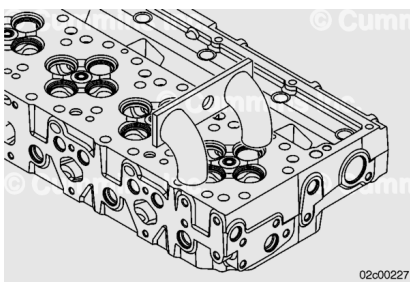


Use a dent puller to remove the expansion plugs.



Use magnetic crack detector, Part Number ST-1166, to find cracks in the combustion face, areas around the valve ports, and the injector bores.

Place the magnetizing head on the combustion face as shown, to check for cracks that run across the cylinder head.



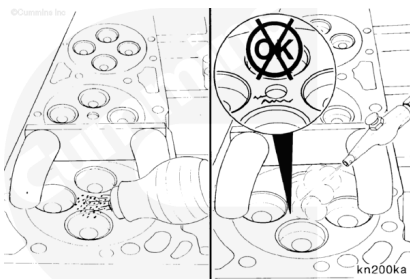
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Fill the powder spray bulb one-third full with metal powder.

Spray the powder lightly onto the combustion face.

Use compressed air to remove the excess metal powder.



Measurements

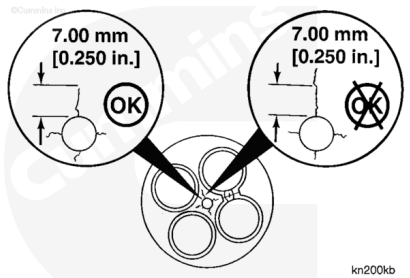
	kpa	psi
Air Pressure	276	40

The powder will remain in the cracks, if present, and show as a white line.

Inspect the cylinder head for cracks in the combustion face.

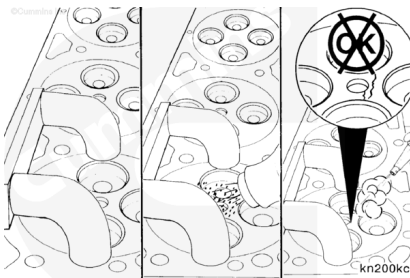
Cracks Around Injector Bore		
mm		in
7.00	MAX	0.276

If any cracks around the injector bore are longer than 7.00 mm [0.276 in], the cylinder head **must** be replaced.



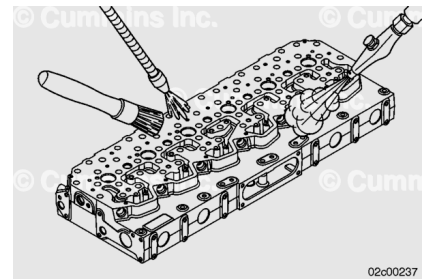
Place the magnetizing head on the combustion face as shown to check for cracks that run lengthwise of the cylinder head.

Repeat the procedure as outlined above.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendation for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Remove all magnetism and use solvent to clean the cylinder head.

Dry with compressed air.

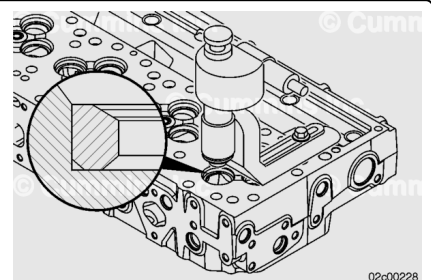
The cylinder head **must** be thoroughly cleaned after using the magnetic crack detector to remove all of the iron fragments.

Assemble

Use exhaust valve seat driver, Part Number 3164097, to drive the exhaust valve seat insert into the counterbore.

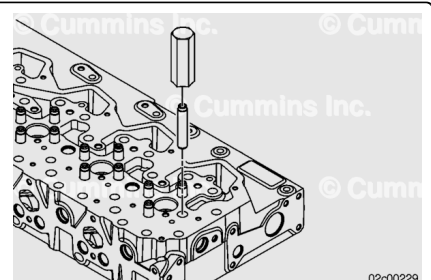
Use intake valve seat driver, Part Number 3164096, to drive the intake valve seat insert into the counterbore.

Make sure the insert is at the bottom of the counterbore.



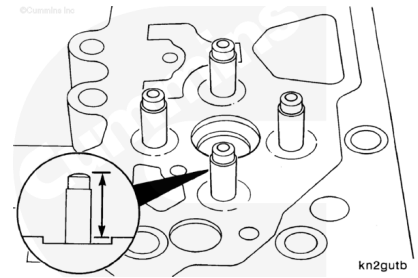
Lubricate the outside diameter of each valve guide with clean 15W-40 engine oil.

Use valve guide driver, Part Number 4919079, to install the new valve guides.



Measure the valve guide installed height above the cylinder head top deck surface.

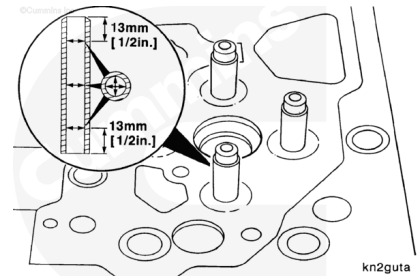
See the Clean and Inspect for Reuse section of this procedure for specifications.



Measure the valve guide inside diameter.

See the Clean and Inspect for Reuse of this procedure for specifications.

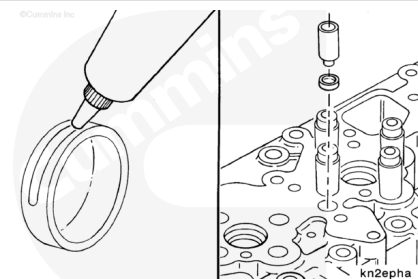
If the valve guide is **not** within the specifications given, the valve guide **must** be replaced.



Apply a ring of Loctite™ sealant, Part Number 150460 or equivalent, to the outside diameter of each expansion plug.

Use expansion plug drivers, Part Numbers 3822372, 3376058, and 3376816, and 4919081 to install the expansion plugs in the cylinder head.

The correct expansion plug driver will install the expansion plugs to the correct depth in the cylinder head.

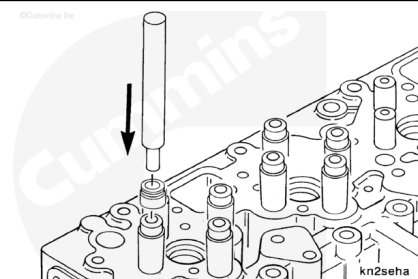


⚠ CAUTION ⚠

Do not use excessive force or the seal will be damaged.

Use valve seal driver, Part Number 4919080, to install the valve seals on the valve guides.

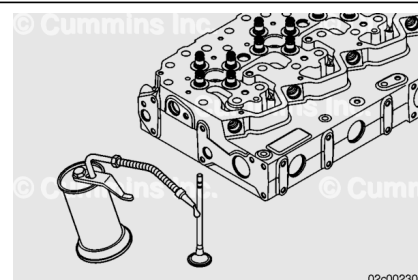
Position the seal on the tool arbor.



Use clean 80W, or higher, gear lube with EP additives to lubricate the valve stems.

Install the valves in the valve guides.

Insert the arbor into the top of the guide and tap the tool with a plastic mallet until the tool seats against the cylinder head surface.



Note : The valve spring guide is **not** assembled at this step.

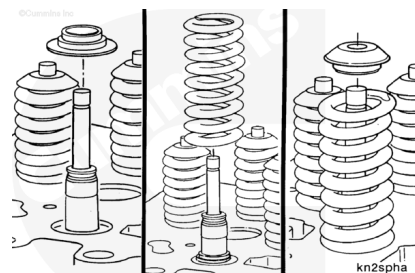
Note : The seal will **not** be flush with the step in the guide.

After the valves are installed, place the cylinder head on a flat surface that will **not** damage the cylinder head surface.

Install the valve spring wear plates.

Install the valve springs.

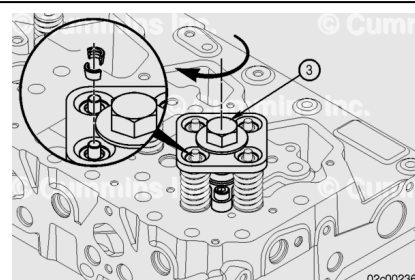
Install the valve spring retainers.



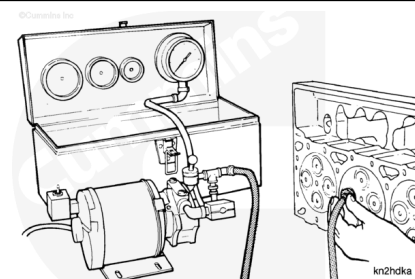
Use valve spring compressor, Part Number 3163940, to compress the valve springs.

Always use new collets when rebuilding the cylinder head.

Install the new valve spring retainer collets.

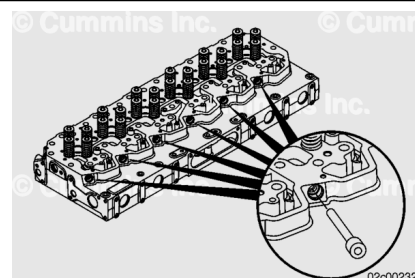


Use vacuum tester, Part Number ST-1257, with vacuum cup, Part Number 3376100, to vacuum test the valve seating. See the Vacuum Test section in this procedure.



Install

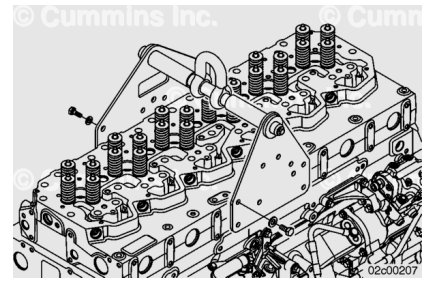
Remove all cylinder head packaging.



Install cylinder head lifting bracket, Part Number 4919361, using the eight M10 capscrews that are included in the cylinder head lifting fixture kit, Part Number 4919360.

Tighten the capscrews.

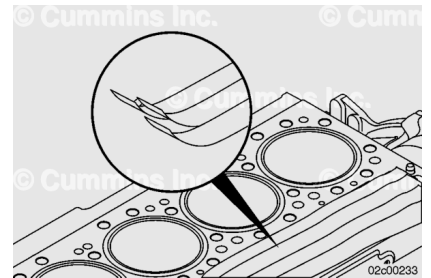
Torque Value: 34 n•m [25 ft-lb]



⚠ CAUTION ⚠

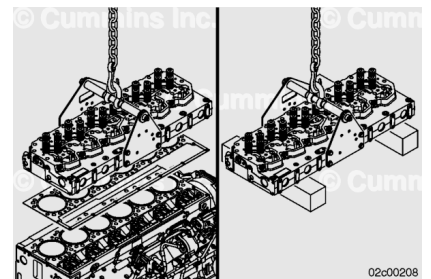
Before the cylinder head is assembled onto the block, all oil passage plugs must be removed. Failure to remove the oil plugs will result in extensive engine damage.

Remove the oil plugs and/or tape from the cylinder head and the cylinder block.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



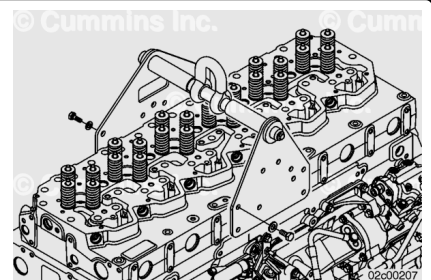
⚠ CAUTION ⚠

Do not drop the cylinder head on the cylinder head gasket. The gasket material can be damaged.

Install the cylinder head gasket.

Use a hoist or hydraulic arm and install the cylinder head.

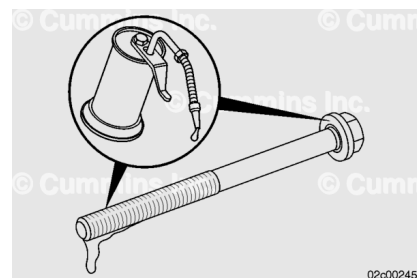
Remove cylinder head lifting bracket, Part Number 4919361.



Use clean 15W-40 oil to lightly coat the cylinder head capscrew threads and bottom of the flange.

Allow the excess oil to drain from the capscrew threads.

Install the 38 M15 cylinder head capscrews and 7 M10 capscrews.



Tighten all the capscrews in the sequence shown to the specified values:

Torque Value:

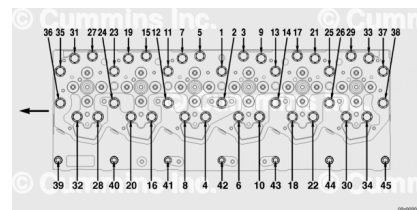
M15 Capscrews

1. 136 n•m [100 ft-lb]
2. 224 n•m [165 ft-lb]
3. Rotate 90 degrees.

Torque Value:

M10 Capscrews 61 n•m [45 ft-lb]

Note : Do not rotate more than two flats.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Install the engine lifting brackets. Refer to Procedure 016-001 in Section 16.
(/qs3/pubsys2/xml/en/procedures/269/269-016-001-tr.html)
- Install all the necessary brackets and clamps.
- Install the fan hub, if applicable. Refer to Procedure 008-036 in Section 8.
(/qs3/pubsys2/xml/en/procedures/269/269-008-036-tr.html)
- Install the air intake manifold. Refer to Procedure 010-023 in Section 10.

(/qs3/pubsys2/xml/en/procedures/269/269-010-023-tr.html)

- Install the fuel drain line. Refer to Procedure 006-013 in Section 6. (/qs3/pubsys2/xml/en/procedures/269/269-006-013-tr.html)
- Install the exhaust manifold. Refer to Procedure 011-007 in Section 11.
(/qs3/pubsys2/xml/en/procedures/269/269-011-007-tr.html)
- Install the rocker lever housing. Refer to Procedure 003-013 in Section 3.
(/qs3/pubsys2/xml/en/procedures/269/269-003-013-tr.html)
- Install the air compressor water outlet tube. Refer to Procedure 012-004 in Section 12.
(/qs3/pubsys2/xml/en/procedures/269/269-012-004-tr.html)
- Install the cooling thermostat housing. Refer to Procedure 008-013 in Section 8.
(/qs3/pubsys2/xml/en/procedures/269/269-008-013.html)
- Install the injectors. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/269/269-006-026-tr.html)
- Install the fuel connectors. Refer to Procedure 006-052 in Section 6. (/qs3/pubsys2/xml/en/procedures/269/269-006-052-tr.html)
- Install the fuel rail. Refer to Procedure 006-060 in Section 6. (/qs3/pubsys2/xml/en/procedures/269/269-006-060-tr.html)
- Install the high-pressure injector supply lines. Refer to Procedure 006-051 in Section 6.
(/qs3/pubsys2/xml/en/procedures/269/269-006-051-tr.html)
- Install the rocker lever assemblies. Refer to Procedure 003-009 in Section 3.
(/qs3/pubsys2/xml/en/procedures/269/269-003-009-tr.html)
- Install the crossheads. Refer to Procedure 002-001 in Section 2. (/qs3/pubsys2/xml/en/procedures/269/269-002-001-tr.html)
- Install the internal actuator harness. Refer to Procedure 019-063 in Section 19.
(/qs3/pubsys2/xml/en/procedures/269/269-019-063-tr.html)

- Install the push rods or push tubes. Refer to Procedure 004-014 in Section 4.
(/qs3/pubsys2/xml/en/procedures/269/269-004-014-tr.html)
- Install the engine brakes, if applicable. Refer to Procedure 020-024 in Section 20.
(/qs3/pubsys2/xml/en/procedures/269/269-020-024-tr.html)
- Install the turbocharger. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/269/269-010-033-tr.html)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/269/269-003-011-tr.html)
- Install the crankcase breather. Refer to Procedure 003-001 in Section 3. (/qs3/pubsys2/xml/en/procedures/269/269-003-001-tr.html)
- Connect the air piping to the intake manifold and turbocharger. Refer to the OEM service manual.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/269/269-008-018-tr.html)
- Connect the batteries. Refer to the OEM service manual.
- Operate the engine to normal operating temperature and check for leaks.